FERMAT'S LITTLE THEOREM

$$a^{p-1} \bmod p \equiv 1$$

Fermat's Little Theorem

If $\gcd(a,p)=1$, p prim, then $a^{p-1} \equiv 1 \mod p$

proof: Consider $S = \{a, 2a, 3a, \dots, (p-1)a, \cancel{pa}\}$

claim: All elements of S are incongruent mod p. In other words their residues mod p form the set $S' = \{\cancel{0}, 1, 2, \dots, p-1\}$

Take $ka, la \in S$, $1 \leq k \neq l \leq p$. Suppose $ka = la \mod p$
$\Rightarrow p \mid (k-l)a \Rightarrow p \mid a$ or $p \mid k-l$ which are both not possible ($\gcd(p,a)=1$, $1 \leq k \neq l \leq p$).

So $S = S'$: $a(2a)(3a)\dots(p-1)a \equiv (p-1)! \mod p$

$\Rightarrow a^{p-1}(p-1)! \equiv (p-1)! \mod p$

$\Rightarrow -a^{p-1} \equiv -1 \mod p \Rightarrow a^{p-1} \equiv 1 \mod p$ $\square$

Zeige $(p-1)! \equiv -1 \mod p$. Bem: $p-1 \equiv -1 \mod p$, $p-2 \equiv -2$. Es gibt in $(p-1)!$ jeweils zwei zueinander inverse Elemente, i.a $g^{-1} \neq g$. Für diese gilt $g \cdot g^{-1} \equiv 1$. Ist $g^{-1} = g$ folgt $g^2 \equiv 1 \Leftrightarrow g \equiv \pm 1 = \begin{cases} 1 \\ p-1 \end{cases}$ $\square$